

of visuospatial processing, attention and perception, information processing, and how we interpret the world around us.

Language and Speech

When attempting to understand the neural bases of language, it is useful to distinguish between grammatical and lexical functions. The grammar–lexicon distinction is different from the more traditional syntax–semantics distinction commonly invoked to improve understanding of the differential effects of brain lesions on language processes (see Chapter 11). *Grammar* is the rule-based system that humans have for ordering words to facilitate communication. For example, in English the typical order of a sentence is subject (noun)–action (verb)–object (noun). The *lexicon* is the mind’s dictionary, where words are associated with specific meanings. The word *dog* is associated with a dog, but so is *Hund*, *kutya*, or *cane*, depending on the language you speak.

The grammar–lexicon distinction takes into account factors such as memory, because with memory, word strings as idioms can be learned by rote. For example, “How are you?” or “*Comment allez-vous?*” is most likely a single lexical entry. Although the lexicon cannot possibly encompass the infinite number of unique phrases and sentences that humans can generate—such as the one you are now reading—memory does play a role in many short phrases. When uttered, such word strings do not reflect an underlying interaction of syntax and semantic systems; they are, instead, essentially an entry from the lexicon.

This grammar–lexicon distinction is more apparent when you are learning a new language. You often learn stock phrases that you speak as a unit rather than struggling with the grammar. It might therefore be predicted that some brain areas ought to be wholly responsible for grammar, whereas the lexicon’s location ought to be more elusive, since it reflects learned information and, thus, is part of the brain’s general memory and knowledge systems. The grammar system, then, ought to be discrete and hence localizable, whereas the lexicon should be distributed and hence more difficult to damage completely.

Language and speech are rarely present in both hemispheres; they are in either one or the other. While the separated left hemisphere normally comprehends all aspects of language, the right hemisphere does have linguistic capabilities, although they are uncommon. Indeed, out of dozens of split-brain patients who have been carefully examined, only six showed clear evidence of residual linguistic functions in the right hemisphere. And even in these patients, the extent of right-hemisphere language functions is severely limited and restricted to the lexical aspects of comprehension.

Interestingly, the left and right lexicons of these special patients can be nearly equal in their capacity, but they are organized quite differently. For example, both hemispheres show a phenomenon called the *word superiority effect* (see Chapter 3). Normal English readers are better able to identify letters (e.g., *L*) in the context of real English words (e.g., *belt*) than when the same letters appear in pseudowords (e.g., *kelt*) or nonsense letter strings (e.g., *ktle*). Because pseudowords and nonwords do not have lexical entries, letters occurring in such strings do not receive the additional processing benefit bestowed on words. Thus, the word superiority effect emerges.

While patients with right-hemisphere language exhibit a visual lexicon, the two hemispheres might access this lexicon in different ways. To test this possibility, investigators used a letter-priming task. Participants were asked to indicate whether a briefly flashed uppercase letter was an *H* or a *T*. On each trial, the uppercase letter was preceded by a lowercase letter that was either an *h* or a *t*. Normally, participants are significantly faster, or primed, when an uppercase *H* is preceded by a lowercase *h* than when it is preceded by a lowercase *t*.

The difference between response latency on compatible (*h*–*H*) versus incompatible (*t*–*H*) trials is taken to be a measure of letter priming. J.W., a split-brain participant, performed a lateralized version of this task in which the prime was displayed for 100 ms to either the right or the left visual field, and 400 ms later the target letter appeared in either the right or the left visual field. The results, shown in **Figure 4.15**, provide no evidence of letter priming for left visual field (LVF) trials, but clear evidence of priming for trials of the right visual field (RVF). Thus, the lack of a priming phenomenon in the disconnected right hemisphere suggests a deficit in letter

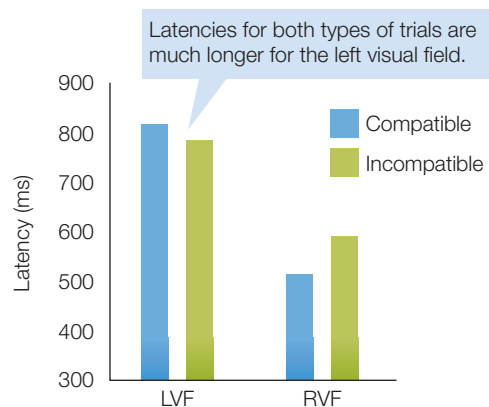


FIGURE 4.15 Letter priming as a function of visual field in split-brain patients.

The graph shows the response latencies for compatible and incompatible pairs of letters in the left and right visual fields (LVF and RVF, respectively). The latencies for both types of trials are much longer for the LVF (right hemisphere).

recognition, prohibiting access to parallel processing mechanisms. J.W. exhibited a variety of other deficiencies in right-hemisphere function as well. For example, he was unable to judge whether one word was superordinate to another (e.g., *furniture* and *chair*), or whether two words were antonyms (e.g., *love* and *hate*).

In sum, there appear to be two lexicons, one in each hemisphere. The right hemisphere's lexicon seems organized differently from the left hemisphere's lexicon, and these lexicons are accessed in different ways. These observations are consistent with the view that lexicons reflect learning processes and, as such, are more widely distributed in the cerebral cortex. It has long been held that in the general population, the lexicon appears to be in the left hemisphere. More recent evidence from functional imaging studies, however, suggests a broader role for the right hemisphere in language processing, although the precise nature of that role remains to be defined (Price, 2012; Vigneau et al., 2011).

Some theorists have suggested that the language ability of the left hemisphere gives it a superior ability to perform higher cognitive functions like making inferences and solving mathematics problems. Split-brain patients who have an extensive right-hemisphere lexicon, however, do not show any attendant increase in their right hemisphere's ability to perform these tasks (Gazzaniga & Smylie, 1984).

In contrast, generative syntax is present in only one hemisphere. Generative syntax means that by following rules of grammar, we can combine words to create an unlimited number of meanings. Although the right hemisphere of some patients clearly has a lexicon, it performs erratically on other aspects of language, such as understanding verbs, pluralizations, the possessive, and active-passive differences. In these patients, the right hemisphere also fails to use word order to disambiguate stimuli for correct meaning. For instance, the meaning of the phrase "The dog chases the cat" cannot be differentiated from the meaning of "The cat chases the dog."

Yet the right hemisphere in these cases can indicate when a sentence ends with a semantically odd word. "The dog chases cat the" would be flagged as wrong. What's more, right hemispheres with language capacities can make grammar judgments. For some peculiar reason, although they cannot use syntax to disambiguate stimuli, they can judge that one set of utterances is grammatical while another set is not. This startling finding suggests that patterns of speech are learned by rote. Yet, recognizing the pattern of acceptable utterances does not mean that a neural system can use this information to understand word strings.

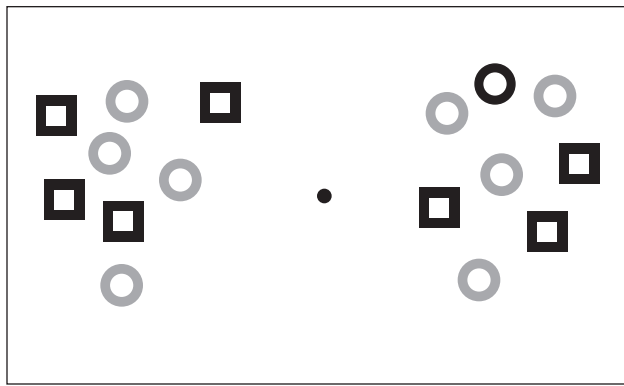
A hallmark of most split-brain patients is that their speech is produced in the left hemisphere and not the

right. This observation, along with amobarbital studies (see Wada & Rasmussen, 1960) and functional imaging studies, confirms that the left hemisphere is the dominant hemisphere for speech production in most (96%) of us. Nonetheless, in a handful of documented cases, split-brain patients can produce speech from both the left and the right hemispheres. Although speech is restricted to the left hemisphere following callosotomy, in this handful of patients the capacity to make one-word utterances from the disconnected right hemisphere has emerged over time.

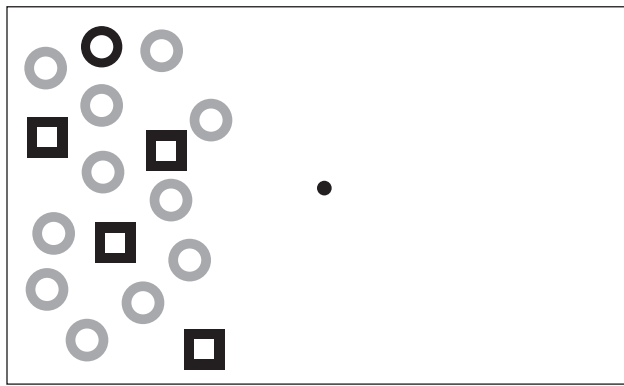
This intriguing development raises the question of whether information is somehow transferring to the dominant hemisphere for speech output or whether the right hemisphere itself is capable of developing speech production. Results from extensive testing supported the latter hypothesis. For example, the patients were able to name an object presented in the left field (say, a spoon) and in the right field (a cow) but were not able to judge whether the two objects were the same. Or, when words like *father* were presented such that the fixation point fell between the *t* and the *h*, the patients said either "fat" or "her," depending on which hemisphere controlled speech production. These findings illustrate that extraordinary plasticity changes can occur, sometimes appearing as late as 10 years after callosotomy. In one patient, in fact, the right hemisphere had no speech production capability for approximately 13 years before it "spoke."

Finally, note that although most language capabilities are left-lateralized, the processing of the emotional content of language appears to be right-lateralized. It is well known that patients with damage to certain regions of the left hemisphere have language comprehension difficulties. Speech, however, can communicate emotional information beyond the meanings and structures of the words. A statement such as "John, come here," can be interpreted in different ways if it is said in an angry tone, a fearful tone, a seductive tone, or a surprised tone.

This nonlinguistic, emotional component of speech is called *emotional prosody*. One patient with left-hemisphere damage reportedly has difficulty comprehending words but shows little deficit in interpreting the meaning of emotional prosody (A. M. Barrett et al., 1999). At the same time, several patients with damage to the temporoparietal lobe in the right hemisphere have been shown to comprehend the meaning of language perfectly but have difficulty interpreting phrases when emotional prosody plays a role (Heilman et al., 1975). This double dissociation between language and emotional prosody in the comprehension of meaning suggests that the right hemisphere is specialized for comprehending emotional expressions of speech.



a



b

FIGURE 4.24 Search array.

(a) A bilateral standard search array and (b) a unilateral guided search array for a black-circle target, where a clue is given: There are fewer black squares than gray circles, in a ratio of about 2:5. The “smart” approach to complete the task faster is to use the clue and concentrate on the black figures. In two out of three split-brain patients, the left hemisphere used the clue, which decreased its reaction time in the guided trials, but the right hemisphere did not.

and gray circles (Figure 4.24). Randomly interspersed through the trials were “guided” trials, where the patient was given a clue: There were fewer black squares than gray circles, in a ratio of about 2:5. A cognitive or “smart” approach would be to use the clue and concentrate on color (black versus gray) instead of on shape (circle versus square), which should enable faster completion of the task.

In two out of three split-brain patients, the left, dominant hemisphere used the clue, which decreased its reaction time in the guided trials, but the right hemisphere did not (Kingstone et al., 1995). In control groups, 70% of people have a faster reaction time in guided trials and use the “smart” strategy. This result indicates that not all people use guided search, but when they do, it’s the left hemisphere doing the work. This apparent discrepancy supports other evidence that multiple mechanisms of attention operate at different stages of visual search processing, from early to late, some of which might be shared across the disconnected hemispheres and others of

which might be independent. *Thus, each hemisphere uses the available resources, but at different stages of processing.*

What’s more, using a “smart strategy” does not mean the left hemisphere is always better at orienting attention. It depends on the job. For instance, the right hemisphere, superior in processing upright faces, automatically shifts attention to where a face is looking, but the left hemisphere does not have the same response to gaze direction (Kingstone et al., 2000).

When thinking about neural resources and their limitations, people often consider the mechanisms that are being engaged while performing voluntary processing. For example, what is happening as you try to rub your stomach, pat your head, and do a calculus problem at the same time? Searching a visual scene, however, calls upon processes that may well be automatic, built-in properties of the visual system itself. Indeed, the hemispheres interact quite differently in how they control reflex and voluntary attentional processes. It appears that reflexive automatic attention orienting is independent in the two hemispheres, as the right hemisphere’s automatic shifting of attention to gaze direction indicates. Voluntary attention orienting, however, is a horse of a different color. Here, it appears, the hemispheres are competing, and the left has more say (Kingstone et al., 1995). That these systems are distinct is reflected in the discovery that splitting brains has a different effect on the processes.

GLOBAL AND LOCAL PROCESSING What does the picture in Figure 4.25 show? A house, right? Now describe it more fully. You might note its architectural style, and you might point out the detailing on the front

**FIGURE 4.25 Global and local representations.**

We represent information at multiple scales. At its most global scale, this drawing is of a house. On a local scale, we can also recognize and focus on the component parts of the house.

door, the double-hung windows running across the front façade, and the shingled roof. Your description of the picture will have been hierarchical. The house can be classified by multiple facets: Its shape and attributes indicate it is a house. But it is also a specific house, with a specific configuration of doors, windows, and materials. This description is hierarchical in that the finer (local) levels of description are embedded in the higher (global) levels. The shape of the house evolves from the configuration of its component parts—an idea that will be developed in Chapter 6.

David Navon (1977) of the University of Haifa introduced a model task for studying **hierarchical structure**. He created stimuli that could be identified on two different levels, as the example in **Figure 4.26** illustrates. At each level, the stimulus contains an identifiable letter. The critical feature is that the letter defined by the global shape is composed of smaller letters (the local shape). In Figure 4.26a, for example, the global *H* is composed of local *F*'s.

Navon was interested in how we perceive hierarchical stimuli. He initially found that the perceptual system first extracted the global shape. The time required to identify the global letter was independent of the identity of the constituent elements, but when it came to identifying the small letters, reaction time was slowed if the global shape was incongruent with the local shapes. Subsequent research qualified these conclusions. Global precedence does depend on object size and the number of

local elements. Perhaps different processing systems are used for representing local and global information.

Lynn Robertson and her colleagues found evidence that supports this hypothesis by studying patients with unilateral brain lesions (L. C. Robertson et al., 1988). When testing such patients, the usual method is to compare the performance of patients who have right-hemisphere lesions against those who have corresponding left-hemisphere lesions. An appealing feature of this approach is that there is no need to lateralize the stimuli to one side or the other; laterality effects are assumed to arise because of the unilateral lesions. For example, if lesions to the left hemisphere result in more disruption in reading tasks, then the deficit is attributed to the hemisphere's specialization in reading processes.

To properly interpret these types of studies, it is necessary to carry out double dissociations (see Chapter 3) to determine whether similar lesions to the opposite hemisphere produce a similar deficit. For instance, it has been demonstrated consistently that lesions in the left hemisphere can produce deficits in language functions (such as speaking and reading) that are not seen in patients with comparable lesions to the right hemisphere. Similarly, lesions to the right hemisphere can disrupt spatial orientation, such as the ability to accurately locate visually presented items. Comparable lesions to the left hemisphere do not cause corresponding spatial deficits.

Robertson presented patients who had a lesion in either the left or the right hemisphere with local and global stimuli in the center of view (the critical laterality factor was whether the lesion was in the left or right hemisphere). Patients with left-side lesions were slow to identify *local* targets, and patients with right-side lesions were slow with *global* targets, demonstrating that the left hemisphere is more adept at representing local information and the right hemisphere is better with global information.

In another study, patients who had recently had a stroke in either the right or the left hemisphere were shown a hierarchical stimulus and asked to reproduce it from memory (Delis et al., 1986). Drawings from patients with left-hemisphere lesions faithfully followed the contour, but without any hint of local elements. In contrast, patients with right-hemisphere lesions produced only local elements (**Figure 4.27**). Note that this pattern was consistent whether the stimuli were linguistic or nonlinguistic; hence, the representational deficits were not restricted to certain stimuli. Note also that, because of the plasticity of the brain, such stark differences might dissipate and not be seen months after the stroke.

Keep in mind that both hemispheres can abstract either level of representation, but they differ in how *efficiently* local and global information are represented. The right is better at the big picture, and the left is more detail oriented.

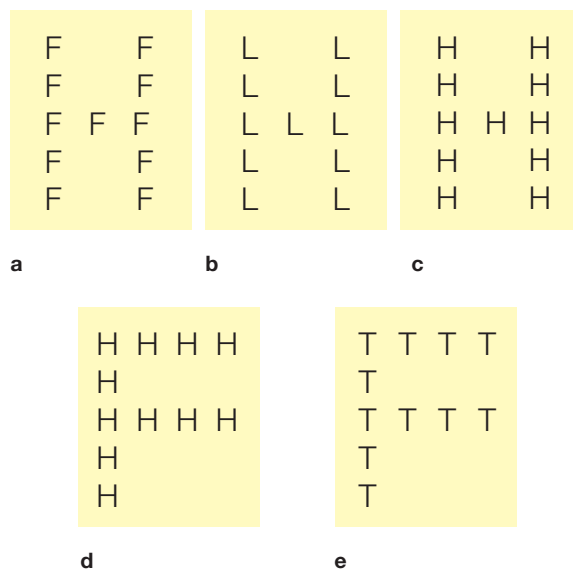


FIGURE 4.26 Local and global stimuli used to investigate hierarchical representation.

Each stimulus is composed of a series of identical letters whose global arrangement forms a larger letter. The participants' task is to indicate whether the stimulus contains an *H* or an *L*. When the stimulus set included competing targets at the two levels (**b**), the participants were instructed to respond either to local targets only or to global targets only. Neither target is present in (**e**).

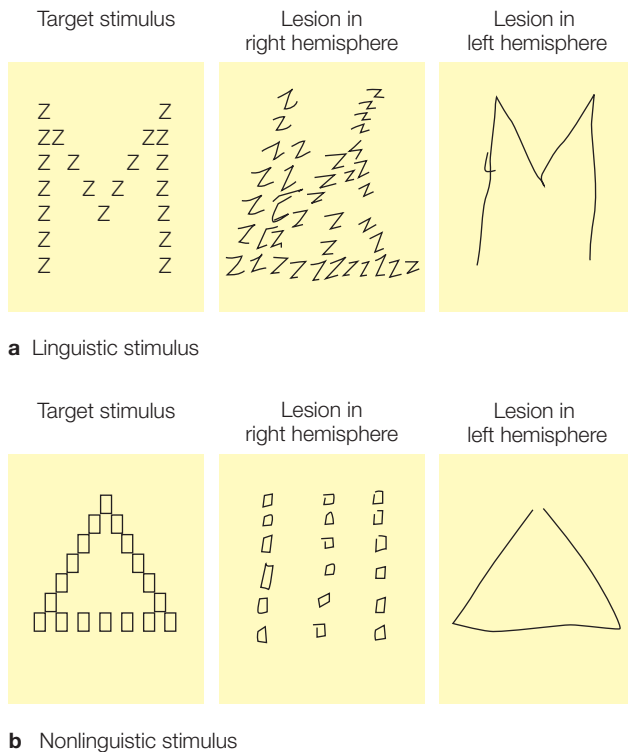


FIGURE 4.27 Extreme failures of hierarchical processing following brain damage.

Two patients were asked to draw the two elements shown in the left column of each panel. The patient with right-hemisphere damage was quite accurate in producing the local element—the Z in (a) or the square in (b)—but failed to arrange these elements into the correct global configuration. The patient with left-hemisphere damage drew the overall shapes—the M in (a) or the triangle in (b)—but left out all of the local elements. Note that for each patient, the drawings were quite consistent for both linguistic (a) and nonlinguistic (b) stimuli, suggesting a task-independent representational deficit.

Thus, patients with left-hemisphere lesions are able to analyze the local structure of a hierarchical stimulus, but they must rely on an intact right hemisphere, which is less efficient at abstracting local information. Further support for this idea comes from studies of local and global stimuli with split-brain patients (L. C. Robertson et al., 1993). Here, too, patients generally identify targets at either level, regardless of the side of presentation. As with normal participants and patients with unilateral lesions, however, split-brain patients are faster at identifying local targets presented to the RVF (the left hemisphere) and global targets presented to the LVF (the right hemisphere).

Theory of Mind

The term *theory of mind* refers to our ability to understand that other individuals have thoughts, beliefs, and desires. In terms of laterality, theory of mind is an interesting case.

You might expect theory of mind to be another hemispheric specialization, lateralized to the left hemisphere like language is, given its dependency on reasoning. Much of the prevailing research on theory of mind, however, suggests that if it is lateralized at all, it is lateralized to the right hemisphere.

Many neuroimaging studies show a network of regions in both hemispheres engaged in theory-of-mind tasks, including the medial portion of the prefrontal cortex (PFC), the posterior portion of the superior temporal sulcus (STS), the precuneus, and the amygdala–temporopolar cortex (**Figure 4.28**). Rebecca Saxe and her colleagues (2009), however, have used a version of the false-belief task (see Chapter 13) in several fMRI studies to demonstrate that the critical component of the theory of mind—the attribution of beliefs to another person—is localized to the temporal–parietal junction in the right hemisphere.

This finding may sound merely interesting to you, but to split-brain researchers it was shocking. Think about it for a second. If this information about the beliefs of others is housed in the right hemisphere, and if, in split-brain patients, it isn't transferred to the speaking, left hemisphere, wouldn't you expect these patients to suffer a disruption in social and moral reasoning? Yet they don't. Split-brain patients act like everyone else. Do these findings also suggest that the recursive nature of thinking about the beliefs of another person is lateralized to the right hemisphere?

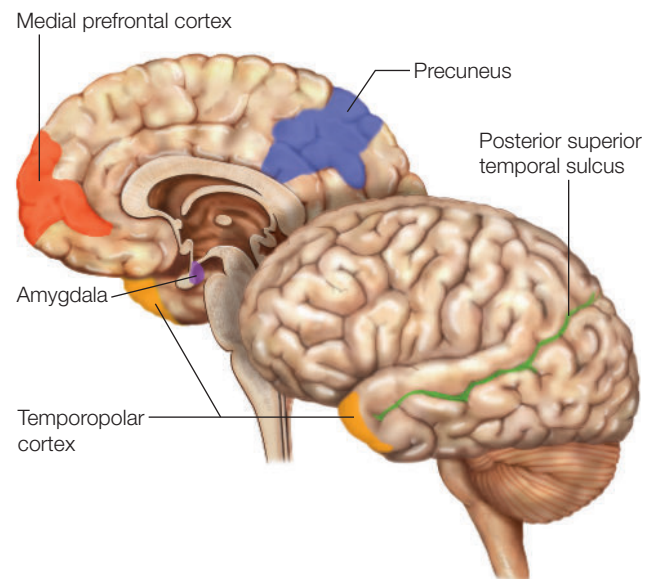


FIGURE 4.28 Theory-of-mind tasks activate a network of regions bilaterally.

These include the medial prefrontal cortex, posterior superior temporal sulcus, precuneus (hidden in the medial longitudinal fissure in the parietal lobe), and the amygdala–temporopolar cortex. The attribution of beliefs is located in the right hemisphere's temporal–parietal junction.

A split-brain study by University of California professor Michael Miller and his colleagues may provide some insight into these questions (M. B. Miller et al., 2010). Miller's team tested three full-callosotomy patients and three partial-callosotomy patients on a moral reasoning task that depended on the ability to attribute beliefs to another person (the same task used by Saxe and colleagues, which produced activations in the right hemisphere).

The task involved hearing a scenario in which the actions of an agent conflicted with the beliefs of the agent. Here's an example: Grace works in a chemical plant, and she is fixing coffee for her friend. She adds a white powder to her friend's coffee, believing that the white powder is sugar. The white powder was mislabeled, however, and is actually quite toxic. Her friend drinks the coffee and dies. After hearing the scenario, the participant is asked this question: Was it morally acceptable for Grace to give the coffee to her friend? Participants with an intact corpus callosum typically say it was morally acceptable to give her friend the coffee, because they think Grace believed that the white powder was sugar and intended no harm. That is, they realize that Grace had a false belief.

If the special mechanisms that attribute belief are lateralized to the right hemisphere, then the speaking left hemisphere of the split-brain patients should be cut off from those mechanisms. Split-brain patients would thus respond in a way that relies on the outcome of the actions (the friend's death) and is not based on the beliefs of the actors. Children younger than age 4 typically respond in this way (because they do not yet have a fully developed theory of mind). Indeed, Miller and colleagues found that all of the split-brain patients responded that Grace's action was morally unacceptable.

This intriguing result leaves open a question: If the left hemispheres of split-brain patients are cut off from this important theory-of-mind mechanism, why don't these patients act like severely autistic individuals, who are unable to comprehend the thinking and beliefs of other people? Some scientists have suggested that the specialized mechanism observed in the right hemisphere may be used for the fast, automatic processing of belief attributions, and that slower, more deliberate reasoning mechanisms of the left hemisphere could perform the same function, given time for deliberation.

In fact, Miller and colleagues observed that patients in the moral reasoning study were often uncomfortable after hearing themselves utter their initial judgments. They would offer spontaneous rationalizations for responding in a particular way. For example, in another scenario a waitress *knowingly* served sesame seeds to somebody who she believed was highly allergic to them. The outcome, however, was harmless, because the person was

not allergic. The split-brain patient judged the waitress's action to be morally acceptable. Some moments later, however, he appeared to rationalize his response by saying, "Sesame seeds are tiny little things. They don't hurt nobody." According to the automatic-versus-deliberate-response hypothesis, the patient had to square his initial response, which did not benefit from information about the belief state of the waitress, with what he rationally and consciously knew was permissible in the world.

There is another interpretation of what could be occurring here, however. When the left hemisphere gives its answer vocally, the right hemisphere hears the judgment for the first time, just as the experimenter does. The right hemisphere's emotional reaction to the judgment may be the same as what the experimenter feels: surprise and dismay. These emotions, produced subcortically, are felt by both hemispheres. Now the left hemisphere experiences an unexpected negative emotional response. What does it do? It has to explain it away. This brings us to a discussion of the left hemisphere's interpreter mechanism.

TAKE-HOME MESSAGES

- The right hemisphere is specialized for detecting upright faces, discriminating among similar faces, and recognizing the faces of familiar others. The left hemisphere is better at recognizing one's own face.
 - Only the left hemisphere can trigger voluntary facial expressions, but both hemispheres can trigger involuntary expressions.
 - Some forms of attention are integrated at the subcortical level, and other forms act independently in the separated hemispheres. Split-brain patients can use either hemisphere to direct attention to positions in either the left or the right visual field.
 - The right hemisphere has limited integrative capacity for cognitive tasks.
 - Functional MRI studies show that the critical component of theory of mind—the attribution of beliefs to another person—is localized to the temporal-parietal junction in the right hemisphere.
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4.4 The Interpreter

A hallmark of human intelligence is our ability to make causal interpretations about the world around us. We make these interpretations on a moment-to-moment basis, usually without realizing it. Imagine going to a movie on a sunny afternoon. Before entering the theater, you notice that the street and parking lot are dry, and only a few clouds are in the sky. When the movie is over and you walk back outside, however, the sky is gray and the

ground is very wet. What do you instantly assume? You probably assume that it rained while you were watching the movie. Even though you did not witness the rain and nobody told you it had rained, you base that causal inference on the evidence of the wet ground and gray skies. This ability to make inferences is a critical component of our intellect, enabling us to formulate hypotheses and predictions about the events and actions in our lives, to create a continuous sensible narrative about our place in the world, and to interpret reasons for the behavior of others.

After a callosotomy, the verbal intelligence and problem-solving skills of a split-brain patient remain relatively intact. There may be minor deficits, including in free recall ability, but for the most part, intelligence remains unchanged. An intact intelligence, however, is true only for the speaking left hemisphere; in the right hemisphere intellectual abilities and problem-solving skills are seriously impoverished. A large part of the right hemisphere's impoverishment can be attributed to the finding that causal inferences and interpretations appear to be a specialized ability of the left hemisphere. So, it is not that the right hemisphere has lost these abilities, but rather that it never had them at all. One of the authors of this book (M.S.G.) has referred to this unique specialization of the left hemisphere as the **interpreter**.

The interpreter has revealed itself in many classic experiments over the years. A typical observation occurs when the speaking left hemisphere offers some kind of rationalization to explain actions that were initiated by the right hemisphere but were spurred on by a motivation unknown to the left hemisphere. For example, when the split-brain patient P.S. was given a command to stand up in a way that only the right hemisphere could view, P.S. stood up. When the experimenter asked him why he was standing, P.S.'s speaking left hemisphere immediately came up with a plausible explanation: "Oh, I felt like getting a Coke." If his corpus callosum had been intact, then P.S. would have responded that he stood up because that was the instruction he had received. One constant finding throughout the testing of split-brain patients is that the left hemisphere *never* admits ignorance about the behavior of the right hemisphere. It *always* makes up a story to fit the behavior.

The interpreter makes itself manifest in a number of ways. Sometimes it interprets the actions initiated by the right hemisphere, as in the example just described, and sometimes it interprets the moods caused by the experiences of the right hemisphere. Emotional states appear to transfer between the hemispheres subcortically, so severing the corpus callosum does not prevent the emotional state of the right hemisphere from being transferred to the left hemisphere, even though all of the perceptions and experiences leading up to that emotional state are still isolated.

Author M.S.G. reported on a case in which he showed some negatively arousing stimuli to the right hemisphere alone. The patient denied seeing anything, but at the same time she was visibly upset. Her left hemisphere felt the autonomic response to the emotional stimulus but had no idea what had caused it. When asked what the matter was, her left hemisphere responded that the experimenter was upsetting her. In this case, the left hemisphere felt the valence of the emotion but was unable to explain the actual cause of it, so the interpreter constructed a theory from the available information.

Probably the most notable example of the interpreter at work is an experiment done by M.S.G. and Joseph LeDoux using a simultaneous concept task (Gazzaniga & LeDoux, 1978). A split-brain patient was shown two pictures, one exclusively to the left hemisphere and one exclusively to the right. Then he was asked to choose, from an array of pictures placed in full view in front of him, those that were associated with pictures lateralized to the left and right sides of the brain (**Figure 4.29**).

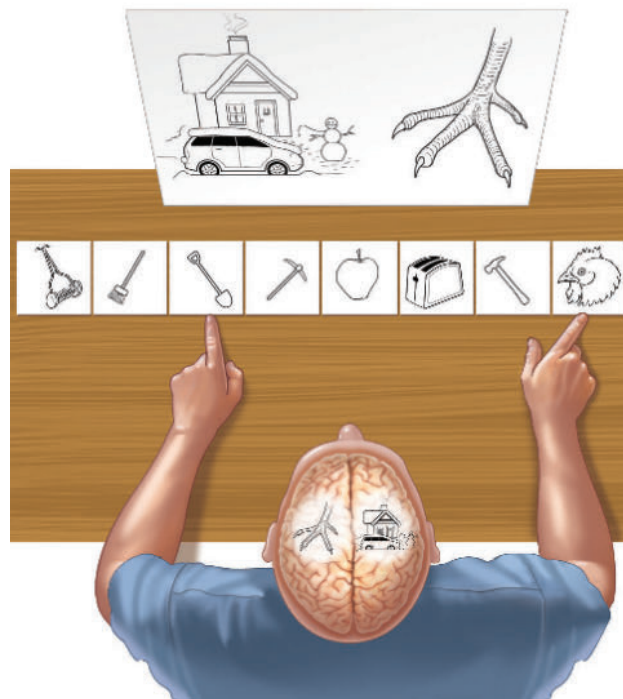


FIGURE 4.29 The interpreter at work.

The left hemisphere of split-brain patient P.S. was shown a chicken claw, and his right hemisphere was shown a snow scene. When P.S. was asked to point to a picture associated with the image he had just seen, his right hand (guided by his left hemisphere) pointed to the chicken, and his left hand pointed to the shovel. When asked why he had pointed to those things, he replied, "Oh, that's simple. The chicken claw goes with the chicken, and you need a shovel to clean out the chicken shed."